# THE HIDDEN CORE OF ROMANIAN: LATENT STRUCTURE OF THE DEX AND GRAUR'S FUNDAMENTAL LEXICAL STOCK

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## Abstract

Dictionaries define words by other words, so every dictionary is circular: a reader who knows none of the defining words can learn nothing from it. Vincent-Lamarre et al. (2016) measured this circularity in English dictionaries by representing a dictionary as a directed graph in which each word is linked to the headwords in whose definitions it occurs. They computed two sets: the Kernel, the part of the dictionary from which all other headwords can be defined (7–12% of headwords), and the minimal grounding set (MinSet), the smallest set of words from which every other headword can be learned through definitions (about 1%).

We apply the method for the first time to Romanian, to the second edition of the DEX (*Dicționarul explicativ al limbii române*) digitised by dexonline (63,843 headwords). Many word forms in definitions are ambiguous (*are*: *avea* 'have' or *ara* 'plough'). A large language model assigns each such form to a headword; on a random sample it agrees with a stronger model on 87% of content-word decisions, and all results are repeated with two procedures that use no model.

The Kernel of the DEX contains 33.2% of the headwords and the MinSet 9.7–9.9%, three to five and about ten times, respectively, the proportions found in English. The main cause is reciprocal definition: 8,761 pairs of headwords occur in each other's definitions (*arbore–copac*, both 'tree'). The Kernel nevertheless contains the basic vocabulary: 98% of the Romanian equivalents of the Swadesh list, a standard list of about 200 basic meanings used in comparative linguistics, and 94.5% of the 1,419 words that Graur (1954) proposed as the fundamental lexical stock of Romanian, against 33.2% expected by chance. The words used most often in definitions, by contrast, are mainly learned French loanwords. The data and code are publicly available.

**Keywords:** lexicography, dictionary graphs, symbol grounding, minimal grounding set, fundamental lexical stock, Romanian, DEX, large language models